

Repair Request Use Case Specification

Performance Classification + Cross-Document Verification

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Supersedes	UC v1.6
Revision Scope	Add performance classification and verification mapping only; actor/flow/state/exception/postcondition semantics are unchanged.
Approval	Business Owner / Project Manager + Architecture/QA review - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

Repair Request Use Case Specification

ข้อกำหนดกรณีใช้งานระบบคำขอซ่อม / MVP Baseline Revision

Document Control	Value
Document ID	UC-RR-001
Version	1.6 - Baseline Revision Candidate
Status	Content Complete - Ready for Baseline Approval
Revision Date	12 September 2026
Approval	Business Owner / Project Manager - formal name/signature pending
Sources	Requirement v1.5; Business Rules v1.5; State RR-STS-001 v1.5
Audience	Business Owner, SA, Developer, Tester, Project Manager
Language	ภาษาไทยเป็นหลัก พร้อมชื่อ Entity / Role / State / API เป็นภาษาอังกฤษ
Purpose	กำหนด Trigger, Actor, Preconditions, Flow, Exception, Validation, Authorization, Audit, Notification, Postcondition and Traceability for each MVP action.

Revision rule: เอกสารฉบับนี้ปิด Audit findings ของชุดวันที่ 11 September 2026 โดยแยกสิ่งที่เป็น Business Baseline ออกจาก Technical Design อย่างชัดเจน. การลงนามอนุมัติจริงยังเป็นหน้าที่ของ Business Owner / Project Manager.

1. Use Case Execution Convention

- Every action validates identity, role, tenant/site scope, ownership/assignment, current state, guard and rowversion at backend.
- Every state-changing action writes append-only Audit History and correlation ID.
- Notification is side effect/outbox; delivery failure logs/retries without rolling back committed business state.
- Validation/authorization/concurrency failure preserves current state and cannot create duplicate transaction.
- Requester/Customer mapping, SLA Risk threshold and MISSED semantics follow D-11..D-15.

2. Detailed Use Cases

UC-RR-001 - Create / Edit Repair Request Draft

Attribute	Specification
Primary Actor	Requester
Supporting Actor	System and downstream role(s) named in flow
Trigger	User chooses New Request / edits own Draft
State Flow	DRAFT -> DRAFT
Preconditions	Authenticated Requester; allowed Site; active masters
Main Flow	Create/edit fields and attachments; server validates; save Draft.
Alternate / Exception	Invalid fields/master/file/scope -> field/access error; no invalid data persisted.
Validation / Authorization	Requester=created_by; Customer scope=tenant_id; rowversion required on edit.
Audit / Notification	Audit create/edit; file scan events as applicable.
Postcondition	DRAFT remains editable by owner.
Traceability	FR-01; BR-01/02/16; ST-RR-001

UC-RR-002 - Submit Request and Handle Duplicate Warning

Attribute	Specification
Primary Actor	Requester
Supporting Actor	System and downstream role(s) named in flow
Trigger	Requester presses Submit
State Flow	DRAFT -> SUBMITTED
Preconditions	Own DRAFT; mandatory data complete
Main Flow	Validate; query D-12 active set; if match warn; require continuation reason; submit; create Request No/SLA/route.
Alternate / Exception	Missing reason/invalid data/stale version -> stay DRAFT. Routing failure -> SUBMITTED, no auto approve.
Validation / Authorization	Server-side scope/master/duplicate + If-Match.
Audit / Notification	Audit submit/reason; notify Approver(s).
Postcondition	SUBMITTED; SLA active.
Traceability	FR-02; BR-02/14/16/17/18; ST-RR-002/003

UC-RR-003 - Route, Approve, Reject or Return for Correction

Attribute	Specification
Primary Actor	Approver
Supporting Actor	System and downstream role(s) named in flow
Trigger	Approval task received
State Flow	SUBMITTED/UNDER_REVIEW -> APPROVED/REJECTED/DRAFT
Preconditions	Assigned/authorized Approver
Main Flow	Review; Approve, Reject(reason) or Return(reason); persist decision.
Alternate / Exception	Unauthorized/stale/invalid state -> deny; routing failure stays SUBMITTED.
Validation / Authorization	Role + route + site scope; reason required on Reject/Return.
Audit / Notification	Audit decision; notify Coordinator or Requester.
Postcondition	Decision state committed once.
Traceability	FR-03; BR-02/04/18; ST-RR-003..006

UC-RR-004 - Cancel Repair Request

Attribute	Specification
Primary Actor	Requester
Supporting Actor	System and downstream role(s) named in flow
Trigger	Requester requests Cancel
State Flow	pre-CONVERTED -> CANCELLED
Preconditions	Own Request; not CONVERTED; reason
Main Flow	Confirm cancellation; commit terminal state; stop SLA if started.
Alternate / Exception	CONVERTED/REJECTED/CANCELLED or no reason -> deny.
Validation / Authorization	Owner/state/reason/If-Match.
Audit / Notification	Audit + NTF-CANCEL.
Postcondition	CANCELLED; no reopen.
Traceability	FR-03; BR-09/12; ST-RR-007

UC-WO-001 - Create Work Order from Approved Request

Attribute	Specification
Primary Actor	Coordinator
Supporting Actor	System and downstream role(s) named in flow
Trigger	Coordinator chooses Convert
State Flow	APPROVED -> CONVERTED; WO OPEN
Preconditions	Request APPROVED; no existing WO
Main Flow	Validate state/version; create WO + link; Request CONVERTED atomically.
Alternate / Exception	Failure/parallel retry -> rollback/no duplicate.
Validation / Authorization	Coordinator/site + unique repair_request_id + If-Match.
Audit / Notification	Audit/outbox; notify operational owner if configured.
Postcondition	Exactly one WO OPEN.
Traceability	FR-04; BR-03; ST-RR-008

UC-WO-002 - View / Track Work Order, Timeline and Audit

Attribute	Specification
Primary Actor	Authorized user
Supporting Actor	System and downstream role(s) named in flow
Trigger	Open list/detail/timeline
State Flow	No state change
Preconditions	Within tenant/site/role/ownership scope
Main Flow	Search/list/detail/timeline with server-side filters/paging.
Alternate / Exception	Out-of-scope guessed ID -> non-leaking deny/not found.
Validation / Authorization	Backend scope authority.
Audit / Notification	Read access may security-audit sensitive access.
Postcondition	Read-only view.
Traceability	FR-09; BR-02/16

UC-WO-003 - Schedule and Assign Service Visit

Attribute	Specification
Primary Actor	Coordinator
Supporting Actor	System and downstream role(s) named in flow
Trigger	Schedule Work Order
State Flow	WO OPEN -> SCHEDULED; new Visit SCHEDULED
Preconditions	Active team/site; valid time
Main Flow	Create Visit; assign team/technician; set schedule.
Alternate / Exception	Invalid/inactive/out of scope -> deny.
Validation / Authorization	Coordinator/site/current WO.
Audit / Notification	Audit schedule/assignment; notify assignee/team lead.
Postcondition	WO/Visit SCHEDULED.
Traceability	FR-05; ST-WO-001; ST-SV-001

UC-WO-005 - Reassign Technician / Team

Attribute	Specification
Primary Actor	Coordinator
Supporting Actor	System and downstream role(s) named in flow
Trigger	Need assignment change
State Flow	Visit SCHEDULED -> SCHEDULED
Preconditions	No Check-in; reason; new active assignee
Main Flow	Validate and update assignment.
Alternate / Exception	After check-in/invalid team -> deny.
Validation / Authorization	Scope/state/reason.
Audit / Notification	Old/new assignee audit; notification.
Postcondition	Visit remains SCHEDULED.
Traceability	FR-05; BR-04; ST-SV-006

UC-WO-006 - Reschedule Service Visit

Attribute	Specification
Primary Actor	Coordinator
Supporting Actor	System and downstream role(s) named in flow
Trigger	Need schedule change
State Flow	SCHEDULED -> RESCHEDULED -> SCHEDULED
Preconditions	No Check-in; reason + valid window
Main Flow	Record old/new, RESCHEDULED then confirm SCHEDULED.
Alternate / Exception	IN_PROGRESS/final/invalid window -> deny.
Validation / Authorization	Scope/state/reason/version.
Audit / Notification	Audit + notify assignee.
Postcondition	New schedule active.
Traceability	FR-05; BR-04; ST-SV-004/005

UC-WO-007 - Cancel Service Visit

Attribute	Specification
Primary Actor	Coordinator
Supporting Actor	System and downstream role(s) named in flow
Trigger	Cancel scheduled Visit
State Flow	SCHEDULED -> CANCELLED
Preconditions	No Check-in; reason
Main Flow	Cancel Visit only; WO/SLA continue unless separately cancelled.
Alternate / Exception	IN_PROGRESS/COMPLETED/MISSED -> deny.
Validation / Authorization	Coordinator/state/reason.
Audit / Notification	Audit + cancellation notification.
Postcondition	Visit terminal CANCELLED.
Traceability	FR-05; BR-09; ST-SV-007

UC-WO-009 - Mark Service Visit as Missed / Unable to Attend

Attribute	Specification
Primary Actor	Coordinator
Supporting Actor	System and downstream role(s) named in flow
Trigger	Visit could not be attended
State Flow	SCHEDULED -> MISSED
Preconditions	No Check-in; missed reason
Main Flow	Mark MISSED; do not auto-create follow-up.
Alternate / Exception	No reason / already started -> deny.
Validation / Authorization	Coordinator/state/reason.
Audit / Notification	Audit + notify Coordinator/Team Lead.
Postcondition	Original MISSED immutable.
Traceability	FR-05; BR-10/19; ST-SV-008

UC-WO-008 - Decide Follow-up after Missed

Attribute	Specification
Primary Actor	Coordinator
Supporting Actor	System and downstream role(s) named in flow
Trigger	MISSED decision task
State Flow	MISSED remains MISSED; optional new Visit SCHEDULED
Preconditions	Original MISSED; decision/reason; valid schedule for new visit
Main Flow	Reschedule/Follow-up/Reassign -> create new linked Visit; Cancel -> record no-follow-up only.
Alternate / Exception	Invalid new schedule/assignee -> no new Visit; original stays MISSED.
Validation / Authorization	Coordinator/site/reason.
Audit / Notification	Audit decision/new visit; notification.
Postcondition	Original immutable; WO continues.
Traceability	FR-05; BR-10/19; ST-SV-009

UC-WO-016 - Check-in and Check-out Work Session

Attribute	Specification
Primary Actor	Assigned Technician
Supporting Actor	System and downstream role(s) named in flow
Trigger	Start/end field work
State Flow	Visit/WO SCHEDULED -> IN_PROGRESS; Session -> CHECKED_OUT; Visit -> COMPLETED
Preconditions	Assigned; no overlap; Check-out facts complete
Main Flow	Check-in opens session; Check-out validates summary/outcome/evidence and closes session/visit.
Alternate / Exception	Overlap/time/order/evidence invalid -> deny; retain current state.
Validation / Authorization	Assignment/state/version.
Audit / Notification	Audit every action.
Postcondition	Completed session/visit; WO remains IN_PROGRESS until summary submit.
Traceability	FR-06; BR-05/06; ST-WS-001/004; ST-SV-002/003

UC-WO-012 - Pause and Resume Work Session

Attribute	Specification
Primary Actor	Assigned Technician
Supporting Actor	System and downstream role(s) named in flow
Trigger	Temporary pause/resume
State Flow	CHECKED_IN <-> PAUSED
Preconditions	Active own session
Main Flow	Pause records pause_start; Resume records resume; enforce sequence.
Alternate / Exception	Double pause/resume or stale -> deny.
Validation / Authorization	Assignment/session/current state.
Audit / Notification	Audit timestamps.
Postcondition	Session valid chronological facts.
Traceability	FR-06; ST-WS-002/003

UC-WO-020 - Submit and Review Work Summary

Attribute	Specification
Primary Actor	Team Lead / Supervisor
Supporting Actor	System and downstream role(s) named in flow
Trigger	Visit work completed
State Flow	WO IN_PROGRESS -> AWAITING_SUPERVISOR_REVIEW -> AWAITING_CUSTOMER_ACCEPTANCE
Preconditions	Checked-out Visit; complete summary/outcome/evidence; designated contact valid before acceptance
Main Flow	Team Lead submits; Supervisor reviews and submits for acceptance.
Alternate / Exception	Missing facts/contact -> remain current state.
Validation / Authorization	Role/site/state.
Audit / Notification	Audit; notify Supervisor then contact.
Postcondition	WO awaiting customer acceptance.
Traceability	FR-07; BR-06/07; ST-WO-003/004

UC-WO-021 - Customer Accept Service Work

Attribute	Specification
Primary Actor	Designated Contact
Supporting Actor	System and downstream role(s) named in flow
Trigger	Acceptance task
State Flow	AWAITING_CUSTOMER_ACCEPTANCE -> COMPLETED
Preconditions	Exact active designated contact, same Site
Main Flow	Review summary/evidence; Accept; create acceptance round.
Alternate / Exception	Other role/contact/stale state -> deny.
Validation / Authorization	Contact identity + state.
Audit / Notification	Audit + notify Supervisor/Coordinator.
Postcondition	WO COMPLETED; awaits Close.
Traceability	FR-07; BR-07/08; ST-WO-005

UC-WO-022 - Customer Reject Service Work

Attribute	Specification
Primary Actor	Designated Contact
Supporting Actor	System and downstream role(s) named in flow
Trigger	Acceptance task
State Flow	AWAITING_CUSTOMER_ACCEPTANCE -> CORRECTIVE_ACTION_REQUIRED
Preconditions	Exact contact; reason required
Main Flow	Reject; create Acceptance(REJECT)+CorrectiveAction(DRAFT) atomically.
Alternate / Exception	No reason/wrong contact/stale -> deny.
Validation / Authorization	Contact/state/reason.
Audit / Notification	Audit + notify Team Lead/Supervisor/Coordinator.
Postcondition	Corrective cycle started.
Traceability	FR-07; BR-07/15; ST-WO-007; ST-CA-001

UC-WO-023 - Create and Approve Corrective Action Plan

Attribute	Specification
Primary Actor	Team Lead / Supervisor
Supporting Actor	System and downstream role(s) named in flow
Trigger	Corrective required
State Flow	CA DRAFT -> PENDING_PLAN_APPROVAL -> APPROVED; WO corresponding corrective states
Preconditions	Plan text + CLEAN evidence
Main Flow	Team Lead submit plan; Supervisor approve.
Alternate / Exception	Missing evidence/unauthorized -> deny.
Validation / Authorization	Role/state/file scan.
Audit / Notification	Audit/notification.
Postcondition	Plan approved for rework.
Traceability	FR-07; BR-15; ST-WO-008/009; ST-CA-002/003

UC-WO-024 - Execute Corrective Work and Resubmit

Attribute	Specification
Primary Actor	Assigned Technician / Team Lead / Supervisor
Supporting Actor	System and downstream role(s) named in flow
Trigger	Approved corrective plan
State Flow	WO CORRECTIVE_PLAN_APPROVED -> IN_PROGRESS -> AWAITING_SUPERVISOR_REVIEW -> AWAITING_CUSTOMER_ACCEPTANCE
Preconditions	Corrective Visit/assignment; evidence required
Main Flow	Schedule corrective Visit; execute work; submit/review summary; same contact re-accepts.
Alternate / Exception	Invalid assignment/evidence -> deny.
Validation / Authorization	Same controls as normal work + same designated contact.
Audit / Notification	Cycle/round audit retained.
Postcondition	New acceptance round pending.
Traceability	FR-07; BR-15; ST-WO-010/003/004

UC-WO-026 - Cancel Work Order

Attribute	Specification
Primary Actor	Supervisor
Supporting Actor	System and downstream role(s) named in flow
Trigger	Need to terminate WO
State Flow	Pre-customer-accept states -> CANCELLED
Preconditions	Before Customer Accept; permission; reason
Main Flow	Cancel WO; cancel non-final Visits; preserve COMPLETED/MISSED history; stop SLA.
Alternate / Exception	After Accept/COMPLETED/CLOSED/CANCELLED -> deny.
Validation / Authorization	Supervisor/state/reason/version.
Audit / Notification	Atomic audit/outbox.
Postcondition	WO CANCELLED terminal.
Traceability	BR-09/12; ST-WO-011

UC-TIME-001 - Request and Approve Work Time Correction

Attribute	Specification
Primary Actor	Technician/Team Lead; Supervisor
Supporting Actor	System and downstream role(s) named in flow
Trigger	Timestamp correction required
State Flow	TC PENDING -> APPROVED/REJECTED; business state unchanged
Preconditions	Supported timestamp; reason; valid sequence
Main Flow	Requester submits old/new/reason; Supervisor decides; approved applies timestamp only.
Alternate / Exception	Unsupported timestamp/invalid sequence/role -> deny.
Validation / Authorization	Role/state + chronological validation.
Audit / Notification	Old/new/reason audit; notify requester.
Postcondition	Traceable corrected work facts; SLA unchanged.
Traceability	FR-08; BR-11/13; ST-TC-*

UC-SLA-001 - Apply Resolution SLA Policy and Authorized SLA Override

Attribute	Specification
Primary Actor	System; Supervisor with permission
Supporting Actor	System and downstream role(s) named in flow
Trigger	Submit / Override
State Flow	SLA event only
Preconditions	MVP policy 24h 24/7; reason for override
Main Flow	On submit snapshot start/due/risk_at; Override updates effective due and risk_at.
Alternate / Exception	Unauthorized/no reason/invalid due -> deny.
Validation / Authorization	Permission; preserve original due/breach.
Audit / Notification	Audit override + notification if threshold crossed.
Postcondition	Traceable SLA record.
Traceability	FR-08; BR-12/13/17; EV-SLA-*

UC-SLA-002 - Monitor SLA Risk and Breach

Attribute	Specification
Primary Actor	System
Supporting Actor	System and downstream role(s) named in flow
Trigger	Clock passes risk/due threshold
State Flow	No business state change
Preconditions	Active SLA not stopped
Main Flow	At T-4h emit Risk once; after due set breach once; terminal stops monitoring.
Alternate / Exception	Notification failure does not alter record/state.
Validation / Authorization	System job scoped by tenant; idempotent event creation.
Audit / Notification	SLA event + recipients.
Postcondition	Risk/breach history retained.
Traceability	FR-08; BR-17/18; EV-SLA-002/003

UC-REP-001 - Search, Timeline and MVP Reports

Attribute	Specification
Primary Actor	Authorized user
Supporting Actor	System and downstream role(s) named in flow
Trigger	Open list/report/export
State Flow	No state change
Preconditions	Authorized scope
Main Flow	Server-side filters/paging; timeline; report areas; export with scope/PII filtering.
Alternate / Exception	Unauthorized/invalid filters -> controlled error.
Validation / Authorization	Tenant/site/role; no client authority.
Audit / Notification	Correlation + security audit as needed.
Postcondition	Read/export result only.
Traceability	FR-09; BR-02/16/18

3. Use Case Coverage Matrix

FR	Use Cases
FR-01	UC-RR-001
FR-02	UC-RR-002
FR-03	UC-RR-003/004
FR-04	UC-WO-001
FR-05	UC-WO-003/005/006/007/008/009
FR-06	UC-WO-012/016; UC-TIME-001
FR-07	UC-WO-020/021/022/023/024/026
FR-08	UC-SLA-001/002; UC-TIME-001
FR-09	UC-WO-002; UC-REP-001

4. Baseline Readiness

All State actions have Use Case coverage and every Use Case references upstream FR/BR/State. No Use Case introduces a new business role/state beyond RR-REQ-001 v1.5 / BR v1.5 / RR-STS-001 v1.5.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีเฉพาะ technical note เดิมเฉพาะเมื่อมีข้อความขัดกัน โดยไม่เปลี่ยน Business Baseline.

5. Use Case Performance Classification & Traceability

Use Case / Path	Class	Performance Design Expectation	Test
UC-WO-002 / Request & WO list/search/timeline	Interactive read	Server-side scope/filter/sort/paging, focused projection, no full aggregate graph for list; bounded query inputs.	TC-PERF-001/002
UC-REP-001 Report/ Export	Reporting	Scope/PII filtering occurs server-side; rows/ bytes/time/memory recorded. Async/streaming threshold is a technical decision from workload evidence, not a new requirement.	TC-PERF-003
UC-RR-001 + file evidence paths	File I/O	Authorized private upload/download; avoid unnecessary full buffering; scan remains asynchronous and only CLEAN is usable.	TC-PERF-005
UC-SLA-002 + notification retry	Background worker	Idempotent detection/delivery from durable DB state; backlog/retry/worker-lag measured.	TC-PERF-004
All state-changing use cases	Command / concurrency	If-Match/current-state guard + atomic transaction; stale/parallel commands cannot create duplicate semantics.	TC-SEC-002 / TC-PERF-006

PERF-DEC-01 - Performance Measurement Statistic (Pending Approval).
Business Baseline ระบุ “Normal List/Search <= 3 seconds at 100 concurrent users and 100,000 records per Site” แต่ไม่ได้ระบุว่า 3 seconds หมายถึง Average, Maximum, p95 หรือ p99. จึง **ห้ามถือ p95 เป็น Requirement ที่อนุมัติแล้ว**. Recommendation สำหรับการอนุมัติคือ p95 end-to-end API response time <= 3.0 seconds; ก่อนอนุมัติ ให้บันทึก Average/p50/p95/p99/throughput/error rate ครบและรายงานผลเป็น diagnostic evidence โดยยังไม่ประกาศ Pass/Fail ของ release gate จาก percentile ใด percentile หนึ่ง.

6. Implementation Boundary

Use Case main/alternate flows remain authoritative for business behavior. Query projection, cancellation, debounce, page-size enforcement, batching and caching are Technical Design choices and must not be inserted into a use case as new business steps unless a Change Request explicitly changes business behavior.